

Statistics in Health Sciences - Assignment 1

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1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set `copd` in the file `copd redux.RData`. Data `copd` is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

“The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data.”

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your `pdf` when compiling your `Rnw`. As a hint, you can follow this procedure:
 1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
 2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
 3. To get the thousands comma separator as in “29,070”, you can use `prettyNum()`,
 4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in $\text{\LaTeX}$ (as seen in Exercise 0).

```
> n_obs <- nrow(copd)
> n_vars <- ncol(copd)
> nans <- 100*colMeans(is.na(copd))
> min_nan <- sprintf("%.2g", min(nans)) # MIRAR
> max_nan <- sprintf("%.2g", max(nans))
> less_5 <- sprintf("%.2g", 100*length(nans[nans<5])/n_vars)
> comp_cases <- sprintf("%.2g", 100*mean(complete.cases(copd)))
> miss_total <- sprintf("%.2g", 100*mean(is.na(copd)))
> total_cells = prettyNum(n_obs*n_vars, big.mark = ",", decimal.mark = ".")
```

“The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 0% to 32%, with 69% of the variables having less than 5% of the data missing. Only 33% of the participants had complete data on all 45 variables. Overall, 5.4% of the information was missing; that is, 5.4% of the total of 15,390 cells ($45 \times 342 = 15,390$) had missing data.”